

ENV 6128

Smart Air Quality Monitoring & Air Pollution Control

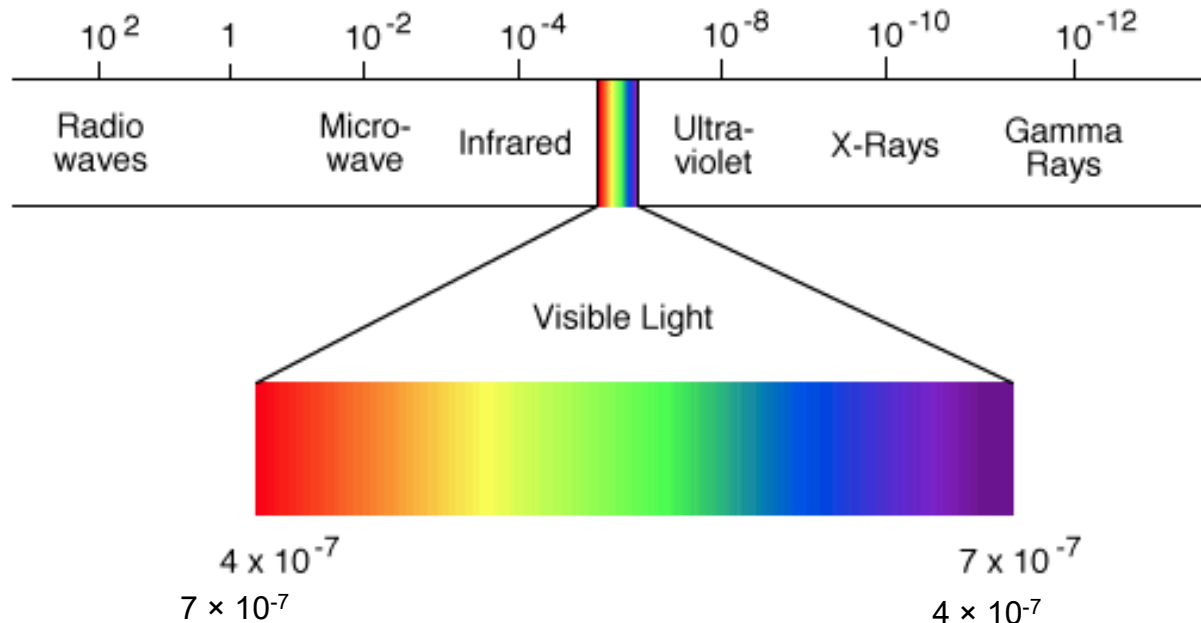
Today's Topics

- Ambient gas pollutant measurement
- Expectation
 - Understand typical measurement technologies for CAPs; low-cost sensor for gaseous pollutants and their limitations

Regulatory Gas CAP measurement

Photometry/Spectrophotometry

- Good for continuous measurements
- Measures the light absorption of a given wavelength range (IR, VIS, UV)



Photometry/Spectrophotometry

- Gases selectively absorb a specific wavelengths
 - IR (0.75 – 100 μm) used primarily for CO
 - VIS (380 – 750 nm) used primarily for NO_2
 - UV (10 – 380 nm) used for O_3 , NO_x , SO_2 , HCHO, and others
- Attenuation of radiation measured to determine absorption



2B tech Model 405 NOx monitor (405 nm)



2B tech Model 205 O_3 monitor (254 nm)

Other detection methods

- Chemiluminescence
 - Chemical reaction used to excite molecules which then luminescence (give off light)
 - Degree of luminescence gives concentration
 - e.g: NO_x and O₃
- UV Fluorescence
 - Similar to photometry but light given off (at a specific wavelength) due to excitation of molecule by UV light
 - Used for SO₂



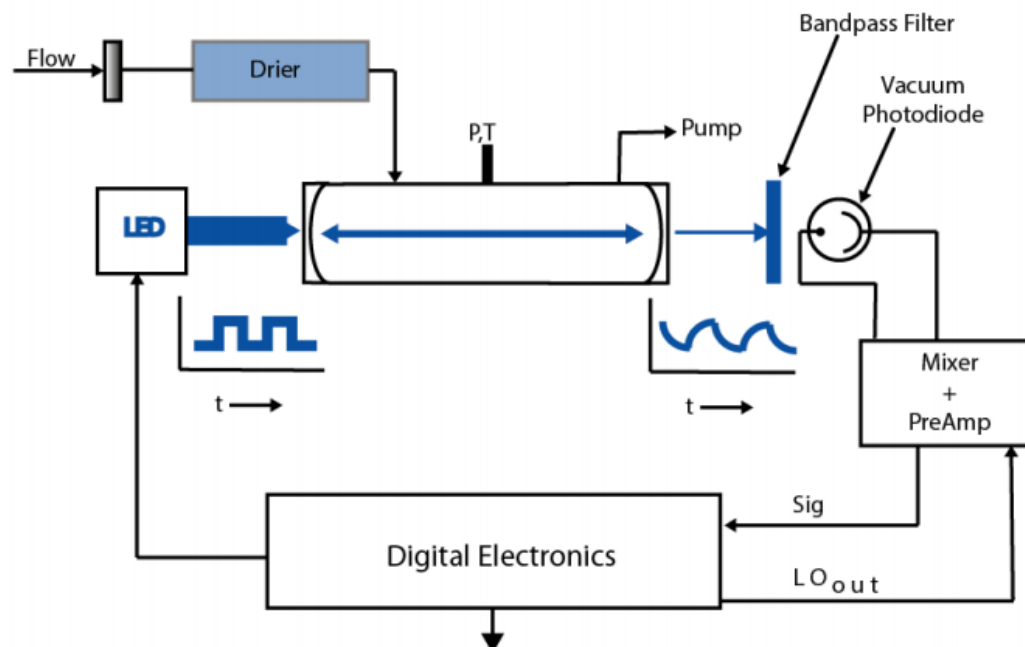
Teledyne Model T200 NO/NO₂/NO_x analyzer



Teledyne Model T100 SO₂ analyzer

Other detection methods

- Cavity Attenuated Phase Shift (CAPS) spectroscopy
 - The presence of gas distort on/off pattern of incoming light
 - Used for NO₂



<https://www.aerodyne.com/wp-content/uploads/2020/02/CAPSNO2.pdf>



Teledyne Model T500U CAPS NO₂ analyzer

Emerging low-cost sensor

Ambient Air Quality Standards

- Regulatory monitoring
 - Use Federal Reference Method (FRM) and Federal Equivalent Method (FEM) instrument
 - Accurate but costly: equipment, power, site, maintenance, data transmission, personal etc
 - Measurement stations limited spatially and temporally
- Pollution concentration could change drastically spatially and temporally
 - More so for primary pollutants, especially traffic-related pollutants such as CO, NO_x

FL monitoring network in FL: <https://floridadep.gov/air/air-monitoring/content/air-quality-today>

Oakland, CA Google Street View vehicle monitoring:

<https://www.edf.org/airqualitymaps/oakland/pollution-and-health-concerns-west-oakland>



Low(er)-cost air quality monitor

- Uses lower-cost sensor to measure pollution
- For particulate matter
 - Mostly optical particle counters
 - Count particle # at limited sizes, convert to mass
- **For gases pollutants**
 - **A wide variety of sensors**
 - **Metal oxides, electrochemical, photochemical, non-dispersive infrared etc**
- Able to measure near-real-time concentration
- Easy integration into “smart” networks



Low-cost gas sensors

- Metal oxide semiconductor (MOS)
 - Conductivity changes when metal oxide surface exposed to target gas
- Electrochemical
 - Redox reaction due to target gas produce measurable current
- Non-Dispersive InfraRed (NDIR)
 - Similar method as described in previous slides



Figaro TGS2611 CH₄ sensor



Alphasense electrochemical sensors



Sensirion SCD30 NDIR CO₂ sensor

More infor read: Jonathan E. Thompson 2016
(<https://doi.org/10.1016/j.teac.2016.06.001>)

Behind the hype

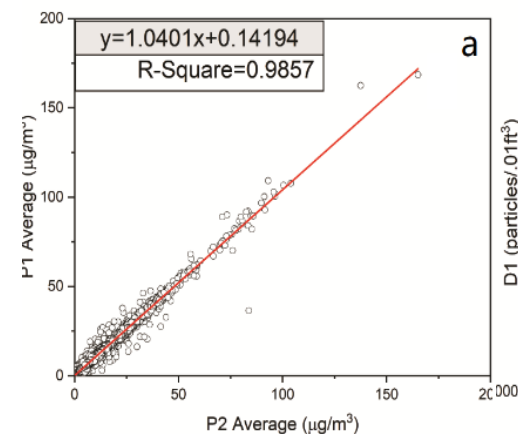
Behind the hype

- Large amount of manufactures rush into market, many with limited understanding on air pollution
- Can we trust the data?
 - Measurement accuracy
 - Inter-unit variability (precision)
 - Environmental impact (temperature, humidity etc)
 - Cross-sensitivity
 - Long-term stability
 - Calibration requirement

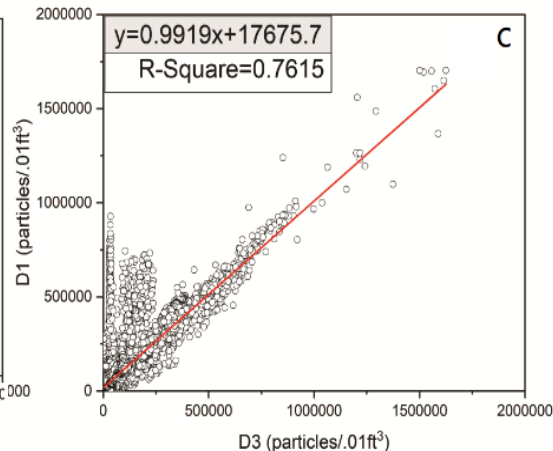
Accuracy, precision and variability

Claimed performance of a
low-cost OPC PM sensor

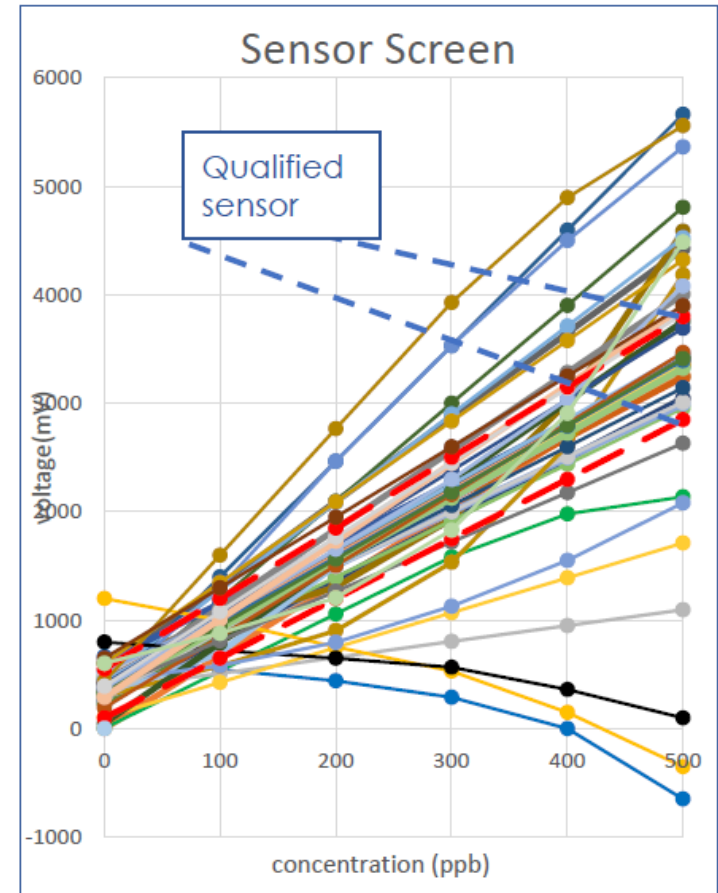
$\pm 10\%$ @ 100 - 500 $\mu\text{g}/\text{m}^3$
 $\pm 10\mu\text{g}$ @ 0 - 100 $\mu\text{g}/\text{m}^3$



Correlations between 2
co-located PM sensors
of brand 1

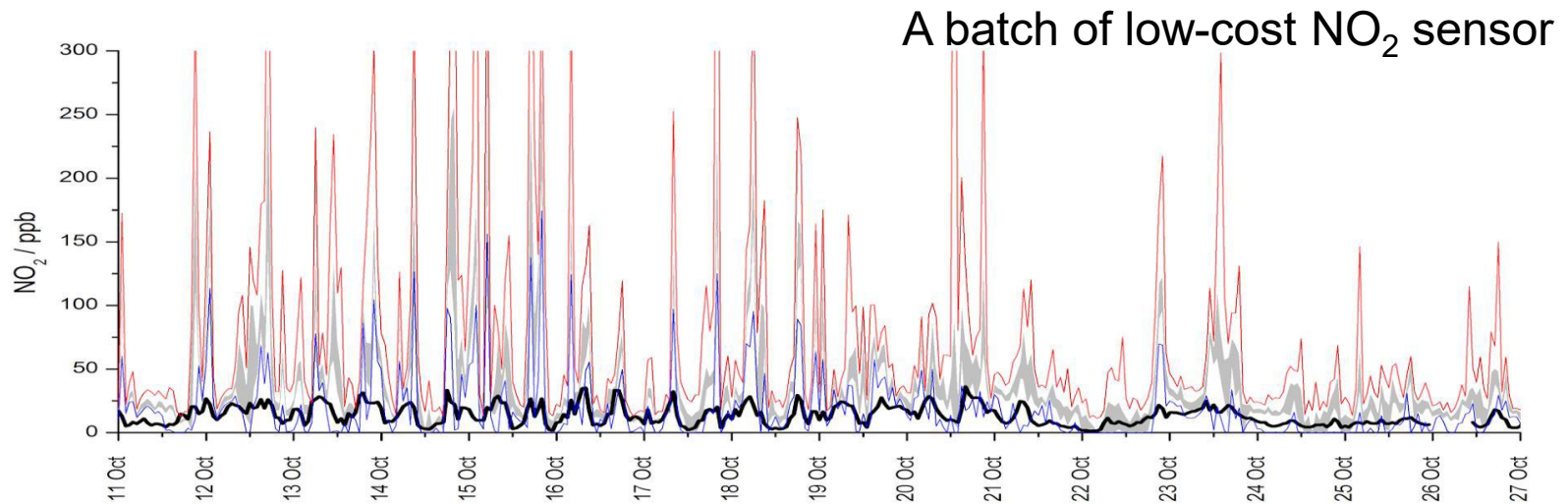
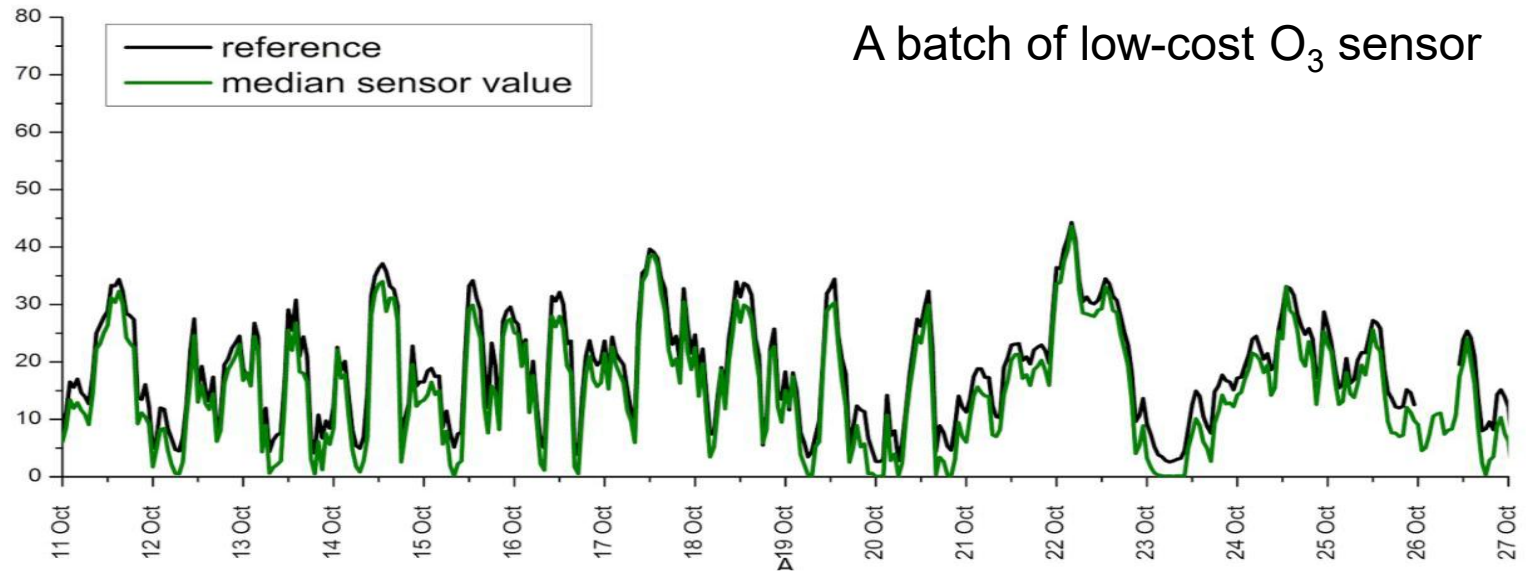


Correlations between 2
co-located PM sensors
of brand 2

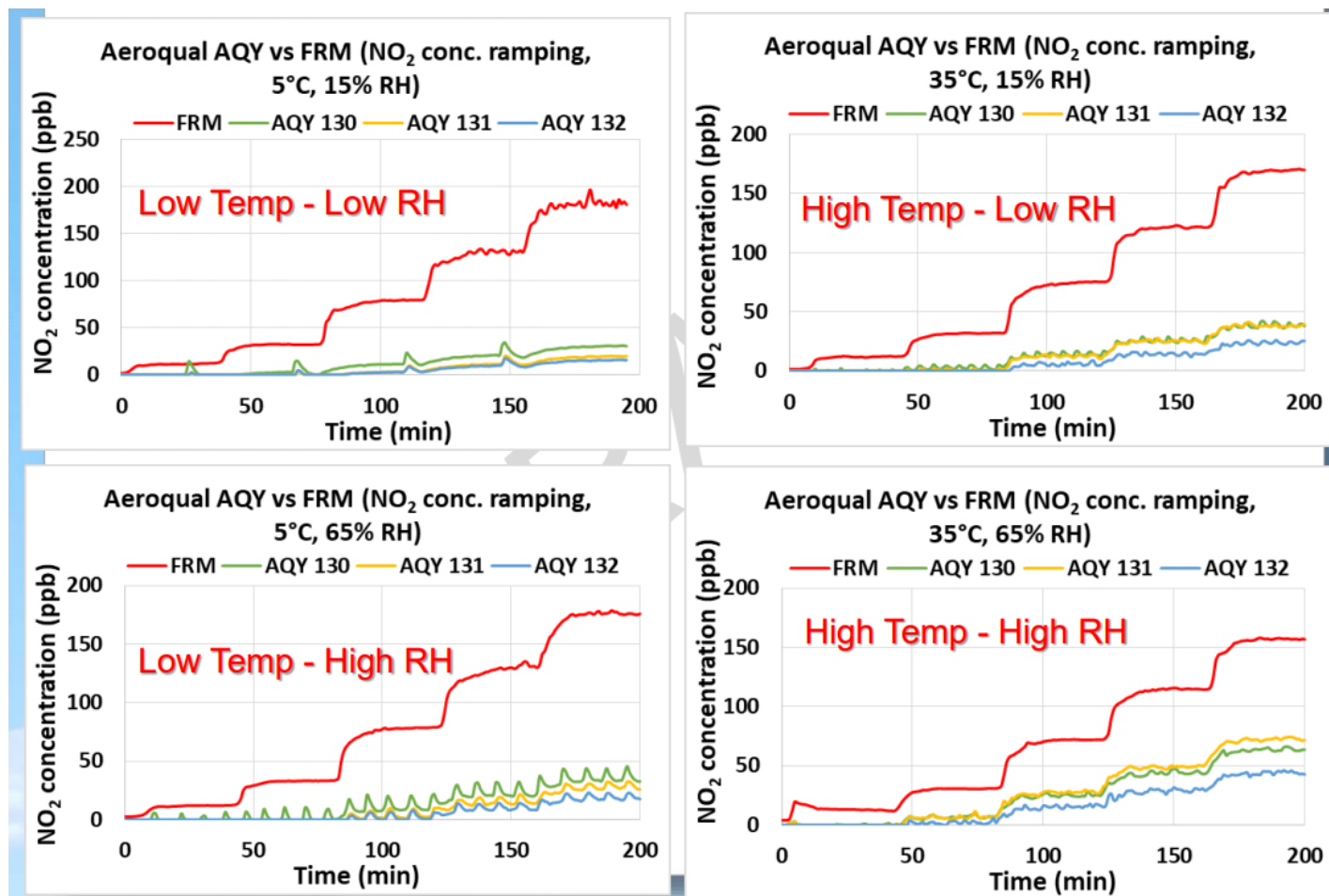


Responses to the same test
gas from gas sensors of the
same brand and same model

Accuracy, precision and variability

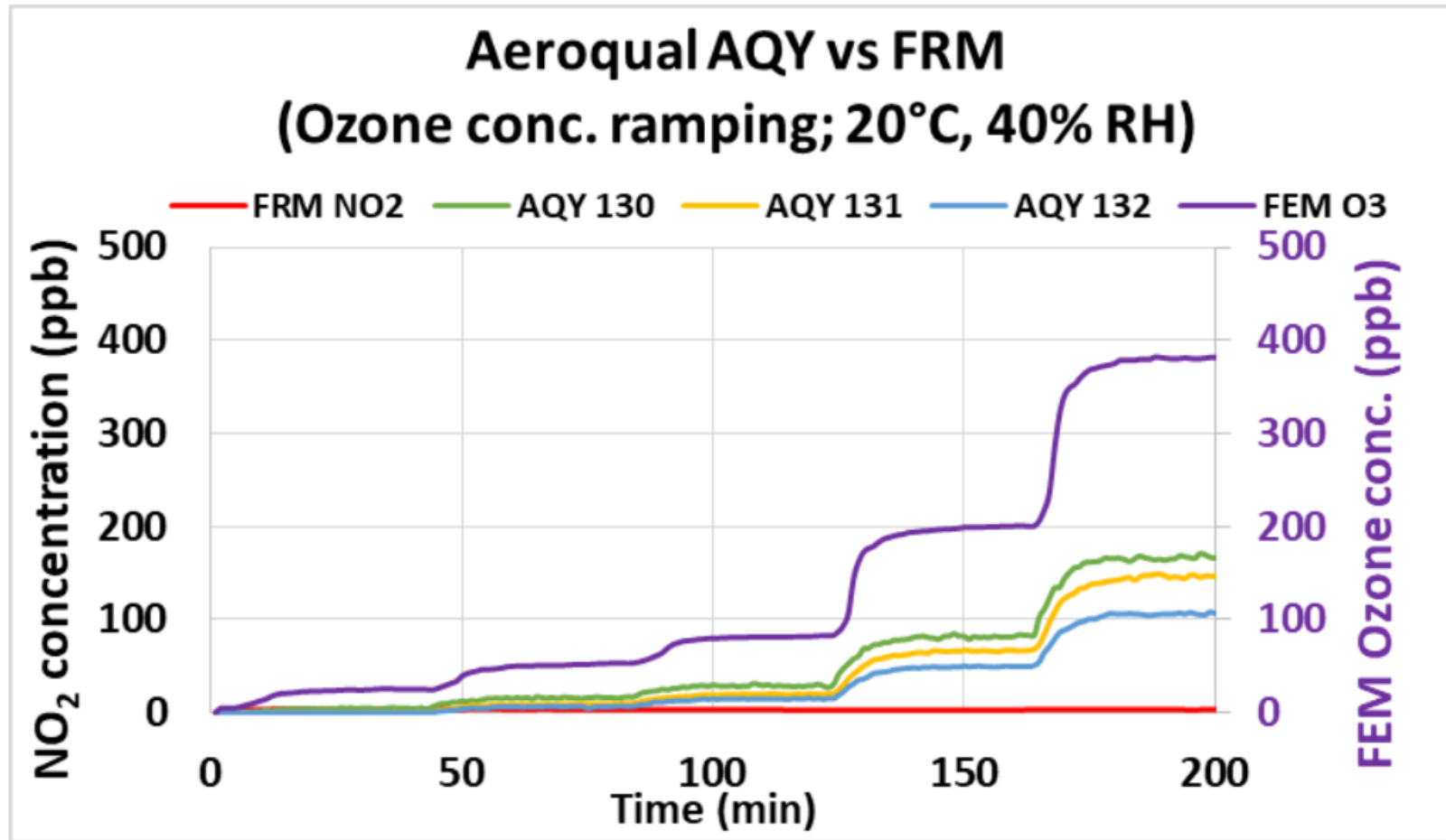


Environmental impacts



Temp and RH impact on NO₂ sensor
From AQ-SPEC evaluation

Cross-sensitivity



Cross-sensitivity of NO₂ sensor with O₃
From AQ-SPEC evaluation

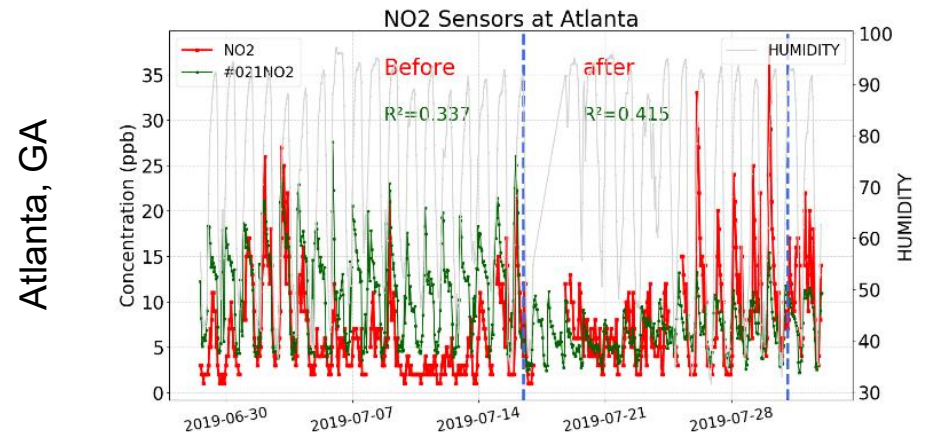
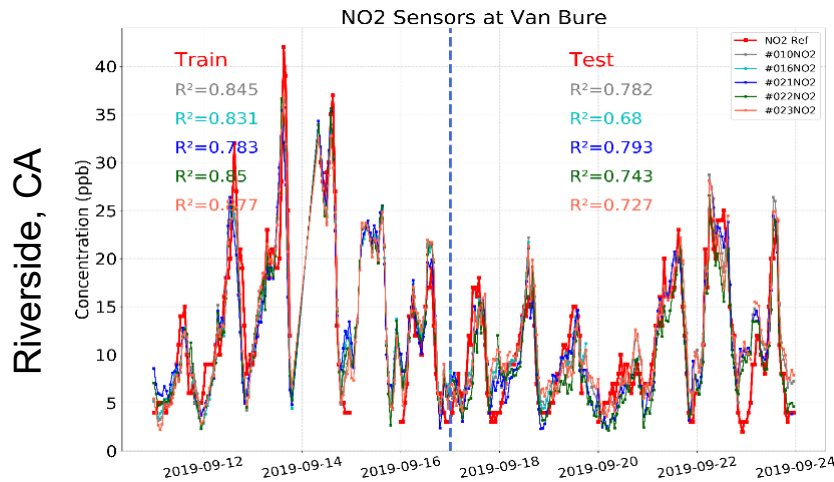
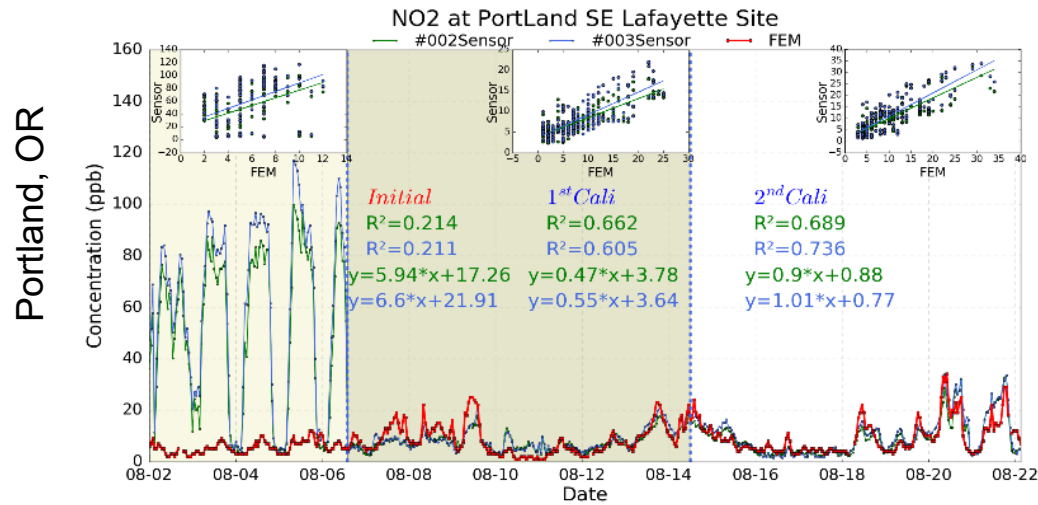
Cross-sensitivity

Sensor	Compound					Co-pollutants		
	CO	SO ₂	NO	O ₃	NO ₂	CO ₂	H ₂	%RH ^a
CO - B4	0.378	-0.013	0.000	0.0200	0.032	0.000	-0.032	0.201
OX-B421	0.000	-0.016	-0.110	0.439	0.44	9.5 x 10 ⁻⁵		0.560
SO ₂ -B4	0.013	0.210	0.023	-0.014	-0.32	9.8 x 10 ⁻⁶		0.000
NO-B4	0	0.007	0.558	-0.011	-0.590	1.8 x 10 ⁻⁵		-0.303
NO ₂ -B4	0	0.004	-0.008	0	0.148	2.3 x 10 ⁻⁵		0.000

Cross-sensitivity of alphasense gas sensors

Lewis et al: https://www.wmo.int/pages/prog/arep/gaw/documents/01_Lewis_LowCostSensors.pdf

Calibration requirement



NO₂

Varying performance of the same model monitor at different locations

Summary

- Smart and connected air quality sensor devices has great potentials to substantially change the landscape of traditional air quality monitoring
- Currently a booming and fast-expanding market, with many devices with varying performances
- Data quality remain a significant concern
- Without careful QA/QC, data collected could be misleading, or even useless
- The vast majority of smart air sensor data can NOT be used for regulatory decision making